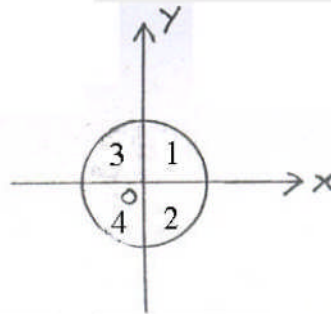




**[SINGLE CORRECT CHOICE TYPE]**

A thin circular disc of radius  $3\pi$  having its centre at the origin is shown in figure. The areal density of each quadrant of the disc is uniform and the relative density of each quadrant is given in the figure. Answer Q.1 to 3.

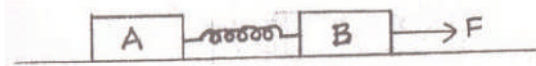


- Q.1 The centre of mass of the semicircular portion (with  $y \geq 0$ ) of the disc with relative densities 1 and 3 is at the point  
(A)  $(-2, 4)$  (B)  $(-4, 2)$  (C)  $(0, 4)$  (D)  $(-4/3, 4)$
- Q.2 The equation of the line passing through the origin and the centre of mass of the disc is  
(A)  $y = x/2$  (B)  $y = 2x$  (C)  $y = -x/2$  (D)  $y = -2x$
- Q.3 Let C be the point at which the centre of mass of the disc is located. A particle of mass equal to that of the disc is to be suitably placed so that the centre of mass of the disc and the particle is now at the point on the rim of the disc closest to C. The position vector  $\vec{r}$  of the particle is  
(A)  $-\frac{2}{\sqrt{5}} (3\pi - 2) (2\hat{i} + \hat{j})$  (B)  $-\frac{2}{\sqrt{5}} [3\pi - (2/\sqrt{5})] (2\hat{i} + \hat{j})$   
(C)  $-\frac{2}{\sqrt{5}} (3\pi - \sqrt{5}) (\hat{i} + 2\hat{j})$  (D) none of these

A man of mass 80 kg is riding on a trolley of mass 40 kg which is rolling along a level surface at a speed of 2 m/s. He jumps off the back of the trolley so that his speed relative to the ground is 1 m/s in the direction opposite to the motion of the trolley. Answer Q.4 to 6.

- Q.4 The speed of the trolley after the man jumps is  
(A)  $4/3$  m/s (B) 4 m/s (C) 8 m/s (D) none of these
- Q.5 The speed of the centre of mass of the system after the man hits the ground and comes to rest is  
(A)  $2/3$  m/s (B)  $8/3$  m/s (C) 4 m/s (D)  $4/3$  m/s
- Q.6 The energy the man expended in jumping is  
(A) zero (B) 427 J (C) 1080 J (D) 1320 J

Two blocks A and B, each of mass 2 kg, are connected by ideal spring and the system is at rest on a frictionless horizontal surface. A constant force  $F = 10 \text{ N}$  is applied on the block B at time  $t = 0$  pulling it away from the other as shown in figure. The elongation caused in the spring at time  $t = 4 \text{ s}$  is 0.2 m. Answer Q.7 to 9.



- Q.7 The displacement of the centre of mass of the system (the blocks and the spring) at  $t = 4 \text{ s}$  has the value  
 (A) 20.4 m (B) 20.2 m (C) 19.8 m (D) different from these
- Q.8 The displacement of the block B at  $t = 4 \text{ s}$  is  
 (A) 20.2 m (B) 20.1 m (C) 20.0 m (D) 19.9 m
- Q.9 The displacement of the block A at  $t = 4 \text{ s}$  is  
 (A) 19.7 m (B) 19.8 m (C) 19.9 m (D) 20.0 m

A particle of mass  $m$  with an initial velocity of  $u \hat{i}$  makes an elastic collision with a particle of mass  $M$  initially at rest. After the collision the particles have velocities which are equal in magnitude but opposite in direction. Answer Q.10 to 13.

- Q.10 The ratio  $M/m$  is  
 (A) 2 (B) 3 (C) 4 (D) none of these

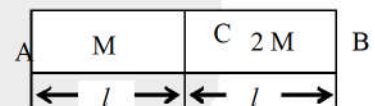
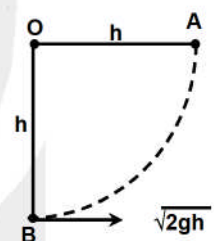
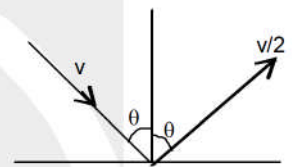
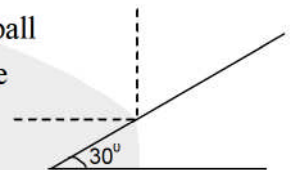
The collision process can be visualized not only in the laboratory frame in which the observer is at rest but also in the centre of mass frame in which the centre of mass is always at rest. Answer Q.11 to 13.

- Q.11 The initial velocity of the mass  $M$  in the centre of mass frame is  
 (A) zero (B)  $+u \hat{i}/4$  (C)  $-u \hat{i}/2$  (D) none of these
- Q.12 The total kinetic energy  $K^*$  of the two particles in the centre of mass frame is  
 (A)  $3mu^2/8$  (B)  $9mu^2/32$  (C)  $mu^2/2$  (D) none of these
- Q.13 If  $K_0$  is the total kinetic energy of the system and  $K^*$  as specified in the previous question,  $(K_0 - K^*)$  is  
 (A) the kinetic energy  $K$  of the centre of mass in the laboratory frame  
 (B)  $-K$  (C)  $2K$  (D) zero

1, 2, 3 and 4 are blocks of at rest on a smooth horizontal surface along a straight line as shown in figure. The blocks 2, 3 and 4 have mass  $m$  each and the block 1 has mass  $2m$ . The block 1 is given a velocity  $u_0 \hat{i}$  so that it collides head-on with the block 2. All collisions are elastic and all those possible have occurred. Answer Q.14 to 17.

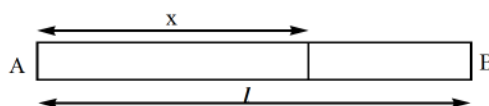
- Q.14 The final velocity of the block 1 is  
 (A)  $u_0 \hat{i}/27$  (B)  $-u_0 \hat{i}/27$  (C)  $u_0 \hat{i}/9$  (D)  $-u_0 \hat{i}/9$
- Q.15 The final velocities of the blocks 2 and 3 are respectively  
 (A)  $u_0 \hat{i}/9$  and  $4u_0 \hat{i}/27$  (B)  $4u_0 \hat{i}/27$  and  $4u_0 \hat{i}/9$   
 (C)  $u_0 \hat{i}/9$  and  $4u_0 \hat{i}/29$  (D)  $4u_0 \hat{i}/9$  and  $4u_0 \hat{i}/3$
- Q.16 The total number of collisions that have occurred is  
 (A) 3 (B) 4 (C) 5 (D) 6

- Q.17 The final velocity of the block 4 relative to the velocity of the centre of mass of the four blocks is  
 (A)  $4u_0 \hat{i}/3$  (B)  $u_0 \hat{i}$  (C)  $14u_0 \hat{i}/15$  (D) none of these
- Q.18 A neutron travelling with a velocity 'v' and kinetic energy 'k' collides perfectly elastically head-on with the nucleus of an atom of mass number 'X' at rest. The fraction of the total energy retained by neutron is  
 (A)  $\frac{(X-1)^2}{(X+1)^2}$  (B)  $\frac{(X+1)^2}{(X-1)^2}$  (C)  $\frac{(X+1)^2}{X}$  (D)  $\frac{(X-1)^2}{X}$
- Q.19 A ball is released from a point, it goes vertically downwards and collides with a smooth inclined plane of angle of inclination of  $30^\circ$  from the ground, then ball goes horizontally. Find the coefficient of restitution between the ball and the inclined plane.
- (A)  $\frac{1}{2}$  (B)  $\frac{1}{3}$  (C) 1 (D) none of these
- Q.20 A ball strikes a rough plane with velocity v and rebounds with velocity v/2. If angle of incidence is equal to angle of reflection then coefficient of friction is
- (A)  $\tan \theta$  (B)  $\frac{1}{2} \tan \theta$  (C)  $\frac{1}{3} \tan \theta$  (D)  $\frac{1}{4} \tan \theta$
- Q.21 Two pendulums A and B having equal lengths and equal bob masses are attached to a common point O and were initially in the position, as shown in the figure. The bob of pendulum A is released from rest and simultaneously that of B was given a horizontal velocity  $\sqrt{2gh}$  as shown in the figure, when the two bobs collide. What will be the ratio of their kinetic energy just before collision?
- (A)  $1:\sqrt{2}$  (B)  $\sqrt{2}:1$  (C)  $1:1$  (D)  $1:2$
- Q.22 The centre of mass of the composite stick made of two uniform stick of masses M and 2M & equal lengths lies at C. Then AC is equal to
- (A)  $3/2 \ell$  (B)  $\ell$   
 (C)  $(4/3) \ell$  (D) None of these

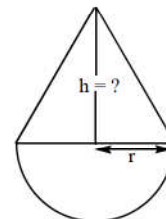


### [SUBJECTIVE TYPE]

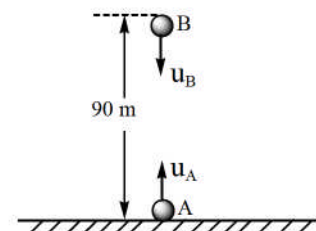
- Q.23 A rod AB of length  $l$  is such that its linear density  $\mu$  varies as  $\mu = a/(b-x)$  where  $x$  is the distance of the section from end A and  $a, b (> l)$  are positive constants of appropriate dimensions. Determine the distance of center of mass of the rod from the end A.



- Q.24 A solid cone is mounted on a solid hemisphere of radius  $r$ , made of same material. What should be the height of the cone, so that the C.G. of the combined mass lies at the centre of base of the cone?



- Q.25 Two particles A and B of mass 1 kg and 2 kg respectively are projected in the directions shown in figure with speeds  $u_A = 200$  m/s and  $u_B = 50$  m/s. Initially they were 90 m apart. Find the maximum height attained by the center of mass of system of the two particles ( $g = 10$  m/s<sup>2</sup>).



### ANSWER KEY

Q.1 A  
Q.5 B  
Q.9 C  
Q.13 A  
Q.17 C

Q.2 A  
Q.6 C  
Q.10 B  
Q.14 A  
Q.18 A

Q.3 B  
Q.7 D  
Q.11 D  
Q.15 B  
Q.19 B

Q.4 C  
Q.8 B  
Q.12 A  
Q.16 D  
Q.20 C

Q.21 C

Q.22 D

$$Q.23 \left[ b - \frac{l}{\ln\left(\frac{b}{b-l}\right)} \right]$$

Q.24  $\sqrt{3}r$

Q.25 115.55 m